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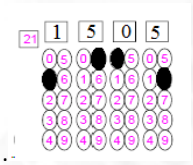
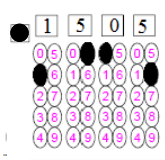
SEC: Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN-Shift-1_Batch-II** **Date: 17-01-2026**

Time: 09.00Am to 12.00Pm **GTM-14** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below

Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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17-01-2026_ Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main-Shift-1_Batch-II_GTM-14_Test Syllabus

MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let $f : \mathbf{R} \to \mathbf{R}$ be a polynomial function of degree four having extreme values at x = 4 and x = 5. If $\lim\limits_{x\to 0}\frac{f(x)}{x^2}=5$, then $f(2)$ is equal to :

1) 12 **2)** 10 **3)** 8 **4)** 14

2. A bag contains 19 unbiased coins and one coin with head on both sides. One coin drawn at random is tossed and head turns up. If the probability that the drawn coin was unbiased, is $\frac{m}{n}, \gcd(m,n)=1$, then n^2-m^2 is equal to:

1) 80 **2)** 60 **3)** 72 **4)** 64

3. Let a random variable X take values 0, 1, 2, 3 with $P(X=0)=P(X=1)=p, P(X=2)=P(X=3)$ and $E(X^2)=2E(X)$. Then the value of 8p – 1 is:

1) 0 **2)** 2 **3)** 1 **4)** 3

4. If the range of the function $f(x)=\frac{5-x}{x^2-3x+2}, x\neq 1,2,$ is $(-\infty,\alpha]\cup[\beta,\infty)$, then $\alpha^2+\beta^2$ is equal to:

1) 190 **2)** 192 **3)** 188 **4)** 194

5. The number of real roots of the equation $x|x-2|+3|x-3|+1=0$ is :

1) 4 **2)** 2 **3)** 1 **4)** 3

6. Let a_n be the nth term of an A. P. If $S_n=a_1+a_2+a_3+\ldots+a_n=700, a_6=7$ and $S_7=7$, then a_n is equal to:

1) 56 **2)** 65 **3)** 64 **4)** 70

7. If the sum of the second, fourth and sixth terms of a G.P. of positive terms is 21 and the sum of its eighth, tenth and twelfth terms is 15309, then the sum of its first nine terms is:

1) 760 **2)** 755 **3)** 750 **4)** 757

8. Let $A=\{(\alpha,\beta)\in \mathbf{R}\times\mathbf{R}:|\alpha-1|\le 4 \text{ and } |\beta-5|\le 6\}$ and $B=\{(\alpha,\beta)\in \mathbf{R}\times\mathbf{R}:16(\alpha-2)^2+9(\beta-6)^2\le 144\}$. Then

1) $B\subset A$ **2)** $A\cup B=\{(x,y):-4\le x\le 4,-1\le y\le 11\}$

3) neither $A\subset B$ nor $B\subset A$ **4)** $A\subset B$

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9. Let the length of a latus rectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ be 10. If its eccentricity is the minimum value of the function $f(t)=t^2+t+\frac{11}{12}$. $t \in R$, then a^2+b^2 is equal to :

1) 125 **2)** 126 **3)** 120 **4)** 115

10. If the area of the region $\left\{(x,y):1+x^2 \le y \le \min\{x+7,11-3x\}\right\}$ is A, then 3A is equal to

1) 50 **2)** 49 **3)** 46 **4)** 47

11. If the locus of $z \in C$, such that $\mathrm{Re}\left(\frac{z-1}{2z+i}\right)+\mathrm{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right)=2,$ is a circle of radius r and center (a, b) then $\frac{15ab}{r^2}$ is equal to:

1) 24 **2)** 12 **3)** 18 **4)** 16

12. Let p be the number of all triangles that can be formed by joining the vertices of a regular polygon P of n sides and q be the number of all quadrilaterals that can be formed by joining the vertices of P. If p + q = 126, then the eccentricity of the ellipse $\frac{x^2}{16}+\frac{y^2}{n}=1$ is:

1) 3/4 **2)** 1/2 **3)** $\frac{\sqrt{7}}{4}$ **4)** $\frac{1}{\sqrt{2}}$

13. Let e_1 and e_2 be the eccentricities of the ellipse $\frac{x^2}{b^2}+\frac{y^2}{25}=1$ and the hyperbola $\frac{x^2}{16}-\frac{y^2}{b^2}=1,$ respectively. If $b < 5$ and $e_1e_2 = 1$, then the eccentricity of the ellipse having its axes along the coordinate axes and passing through all four foci (two of the ellipse and two of the hyperbola) is:

1) 4/5 **2)** 3/5 **3)** $\frac{\sqrt{7}}{4}$ **4)** $\frac{\sqrt{3}}{2}$

14. Let the system of equations

$x+5y-z=1$

$4x+3y-3z=7$

$24x+y+\lambda z=\mu$

$\lambda, \mu \in R$, have infinitely many solutions. Then the number of the solutions of this system, If x, y, z are integers and satisfy $7 \le x+y+z \le 77,$ is :

1) 3 **2)** 6 **3)** 5 **4)** 4

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15. Let y = y(x) be the solution of the differential equation

$\left(x^2+1\right)y'-2xy=\left(x^4+2x^2+1\right)cosx, y(0)=1$. Then $\int_{-3}^{3} y(x)dx$ is :

1) 24 **2)** 36 **3)** 30 **4)** 18

16. Consider the lines L_1 : x –1 = y–2 = z and L_2 : x –2 =y=z–1. Let the feet of the perpendiculars from the point P(5,1,-3) on the lines L_1 and L_2 be Q and R respectively. If the area of the triangle PQR is A, then $4A^2$ is equal to :

1) 139 **2)** 147 **3)** 151 **4)** 143

17. Let $\vec{a}$ and $\vec{b}$ be the vectors of the same magnitude such that $\frac{|\vec{a}+\vec{b}|+|\vec{a}-\vec{b}|}{|\vec{a}+\vec{b}|-|\vec{a}-\vec{b}|}=\sqrt{2}+1$. Then $\frac{|\vec{a}+\vec{b}|^2}{|\vec{a}|^2}$ is:

1) $2+4\sqrt{2}$ **2)** $1+\sqrt{2}$ **3)** $2+\sqrt{2}$ **4)** $4+2\sqrt{2}$

18. If the equation of the line passing through the point $\left(0,-\frac{1}{2},0\right)$ and perpendicular to the lines $\vec{r}=\lambda\left(\hat{i}+a\hat{j}+b\hat{k}\right)$ and $\vec{r}=\left(\hat{i}-\hat{j}-6\hat{k}\right)+\mu\left(-b\hat{i}+a\hat{j}+5\hat{k}\right)$ is, $\frac{x-1}{-2}=\frac{y+4}{d}=\frac{z-c}{-4}$, then $a+b+c+d$ is equal to :

1) 10 **2)** 14 **3)** 13 **4)** 12

19. If the orthocenter of the triangle formed by the lines y = x + 1, y = 4x – 8 and y = mx + c is at (3, –1), then m – c is:

1) 0 **2)** -2 **3)** 4 **4)** 2

20. The number of solutions of the equation $\cos 2\theta\cos\frac{\theta}{2}+\cos\frac{5\theta}{2}=2\cos^3\frac{5\theta}{2}$ in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ is:

1) 7 **2)** 5 **3)** 6 **4)** 9

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SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. If $\int\left(\frac{1}{x}+\frac{1}{x^3}\right)\left(\sqrt[23]{3x^{-24}+x^{-26}}\right)dx=-\frac{\alpha}{3(\alpha+1)}\left(3x^{\beta}+x^{\gamma}\right)^{\frac{\alpha+1}{\alpha}}+C, x>0,\ (\alpha,\beta,\gamma\in Z)$, Where C is the constant of integration. Then $\alpha+\beta+\gamma$ is equal to _____.

22. For $t>-1$, let α_t and β_t be the roots of the equation $\left((t+2)^{\frac{1}{7}}-1\right)x^2+\left((t+2)^{\frac{1}{6}}-1\right)x+\left((t+2)^{\frac{1}{21}}-1\right)=0$. If $\lim\limits_{t\to-1^+}\alpha_t=a$ and $\lim\limits_{t\to-1^+}\beta_t=b$, then $72(a+b)^2$ is equal to _____.

23. The sum of the series $2\times1\times{}^{20}C_4-3\times2\times{}^{20}C_5+4\times3\times{}^{20}C_6-5\times4\times{}^{20}C_7+\ldots+18\times17\times{}^{20}C_{20}$, is equal to

24. Let the lengths of the transverse and conjugate axes of a hyperbola in standard form be 2a and 2b, respectively, and one focus and the corresponding directrix of this hyperbola be (–5,0) and 5x + 9 = 0, respectively. If the product of the focal distances of a point $\left(\alpha,2\sqrt{5}\right)$ on the hyperbola is p, then 4p is equal to ____

25. If the function $f(x)=\frac{\tan(\tan x)-\sin(\sin x)}{\tan x-\sin x}$ is continuous at x = 0, then f(0) is equal to _____.

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. The temperature of a gas having 2.0×10^{25} molecules per cubic meter at 1.38 atm (Given, $k=1.38\times10^{-23}JK^{-1}$)) is :

1) 500 K **2)** 200 K **3)** 100 K **4)** 300 K

27. A stone of mass 900g is tied to a string and moved in a vertical circle of radius 1m making 10 rpm. The tension in the string, when the stone is at the lowest point is $(\text{if } \pi^2=9.8 \text{ and } g=9.8 m/s^2)$:

1) 97 N **2)** 9.8 N **3)** 8.82 N **4)** 17.8 N

28. The bob of a pendulum was released from a horizontal position. The length of the pendulum is 10m. If it dissipates 10% of its initial energy against air resistance, the speed with which the bob arrives at the lowest point is : [Use,g:$10ms^{-2}$]

1) $6\sqrt{5}ms^{-1}$ **2)** $5\sqrt{6}ms^{-1}$ **3)** $5\sqrt{5}ms^{-1}$ **4)** $2\sqrt{5}ms^{-1}$

29. If the distance between object and its two times magnified virtual image produced by a curved mirror is 15 cm, the focal length of the mirror must be :

1) 15 cm **2)** –12 cm **3)** –10 cm **4)** 10/3 cm

30. In Young's double slit experiment, light from two identical sources are superimposing on a screen. The path difference between the two lights reaching at a point on the screen is $\frac{7\lambda}{4}$. The ratio of intensity of fringe at this point with respect to the maximum intensity of the fringe is :

1) 1/2 **2)** 3/4 **3)** 1/3 **4)** 1/4

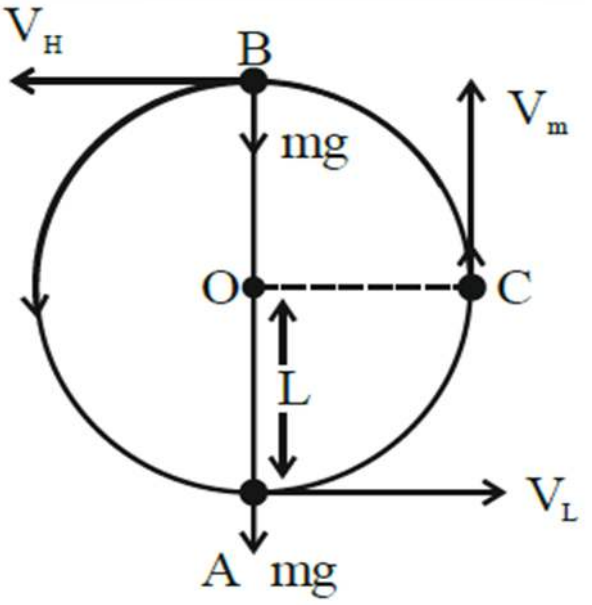
31. A small liquid drop of radius R is divided into 27 identical liquid drops. If the surface tension is T, then the work done in the process will be :

1) $8\pi R^2T$ **2)** $3\pi R^2T$ **3)** $\frac{1}{8}\pi R^2T$ **4)** $4\pi R^2T$

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32. A bob of mass 'm' is suspended by a light string of length 'L'. It is imparted a minimum horizontal velocity at the lowest point A such that it just completes half circle reaching the top most position B. The ratio of kinetic energies $\frac{(K.E.)_A}{(K.E.)_B}$ is :



1) 3 : 2 **2)** 5 : 1 **3)** 2 : 5 **4)** 1 : 5

33. A wire of length L and radius r is clamped at one end. If its other end is pulled by a force F, its length increases by *l*. If the radius of the wire and the applied force both are reduced to half of their original values keeping original length constant, the increase in length will become:

1) 3 times **2)** 3/2 times **3)** 4 times **4)** 2 times

34. A planet takes 200 days to complete one revolution around the Sun. If the distance of the planet from Sun is reduced to one fourth of the original distance, how many days will it take to complete one revolution :

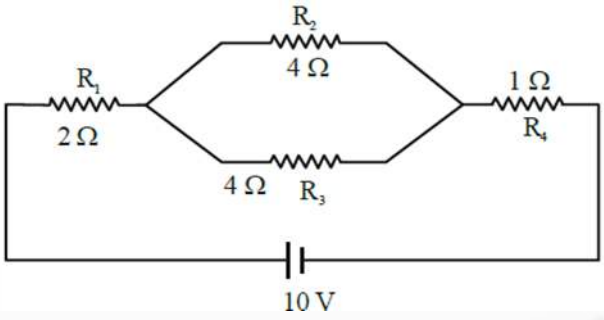
1) 25 **2)** 50 **3)** 100 **4)** 20

35. A plane electromagnetic wave of frequency 35 MHz travels in free space along the X-direction. At a particular point (in space and time) $\vec{E} = 9.6\hat{j}V/m$. The value of magnetic field at this point is :

1) $3.2\times10^{-8}\hat{k}T$ **2)** $3.2\times10^{-8}\hat{i}T$ **3)** $9.6\hat{j}T$ **4)** $9.6\times10^{-8}\hat{k}T$

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36. In the given circuit, the current in resistance R_3 is:



1) 1 A **2)** 1.5 A **3)** 2 A **4)** 2.5 A

37. An electric field is given by $\left(6\hat{i}+5\hat{j}+3\hat{k}\right)N/C$. The electric flux through a surface area $30\hat{i}\,m^2$ lying in YZ-plane (in SI unit) is :

1) 90 **2)** 150 **3)** 180 **4)** 60

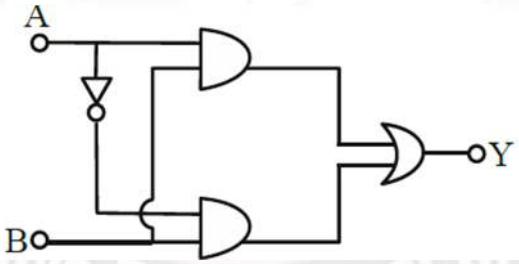
38. Two particles X and Y having equal charges are being accelerated through the same potential difference. Thereafter they enter normally in a region of uniform magnetic field and describes circular paths of radii R_1 and R_2 respectively. The mass ratio of X and Y is:

1) $\left(\frac{R_2}{R_1}\right)^2$ **2)** $\left(\frac{R_1}{R_2}\right)^2$ **3)** $\left(\frac{R_1}{R_2}\right)$ **4)** $\left(\frac{R_2}{R_1}\right)$

39. Two sources of light emit with a power of 200 W. The ratio of number of photons of visible light emitted by each source having wavelengths 300 nm and 500 nm respectively, will be:

1) 1 : 5 **2)** 1 : 3 **3)** 5 : 3 **4)** 3 : 5

40. The truth table for this given circuit is:



1)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

2)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

3)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

4)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

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41. A physical quantity Q is found to depend on quantities a, b, c by the relation $Q=\frac{a^4b^3}{c^2}$. The percentage error in a, b and c are 3%, 4% and 5% respectively. Then, the percentage error in Q is:

1) 66% **2)** 43% **3)** 34% **4)** 14%

42. In an a.c. circuit, voltage and current are given by:

V = 100 sin (100 t)V and $I=100\sin\left(100t+\frac{\pi}{3}\right)$ mA respectively. The average power dissipated in one cycle is:

1) 5 W **2)** 10 W **3)** 2.5 W **4)** 25 W

43. Given below are two statements :

Statement I : Most of the mass of the atom and all its positive charge are concentrated in a tiny nucleus and the electrons revolve around it, is Rutherford's model.

Statement II : An atom is a spherical cloud of positive charges with electrons embedded in it, is a special case of Rutherford's model.

In the light of the above statements, choose the most appropriate from the options given below.

1) Both statement I and statement II are false.

2) Statement I is false but statement II is true.

3) Statement I is true but statement II is false.

4) Both statement I and statement II are true.

44. A particle is moving in a straight line. The variation of position 'x' as a function of time 't' is given as $x=\left(t^3-6t^2+20t+15\right)m$. The velocity of the body when its acceleration becomes zero is

1) 6 m/s **2)** 10 m/s **3)** 8 m/s **4)** 4 m/s

45. *N* moles of a polyatomic gas (f = 6) must be mixed with two moles of a monoatomic gas so that the mixture behaves as a diatomic gas. The value of *N* is:

1) 4 **2)** 3 **3)** 6 **4)** 2

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

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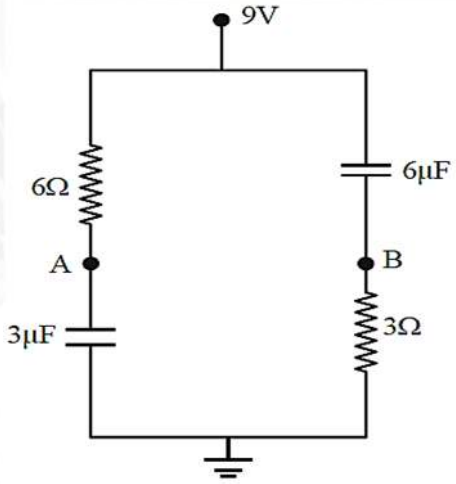

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46. Two metallic wires P and Q have same volume and are made up of same material. If their area of cross sections are in the ratio 4:1 and force F_1 is applied to P, an extension of Δl is produced. The force which is required to produce same extension in Q is F_2. The value of $\frac{F_1}{F_2}$ is_____ .

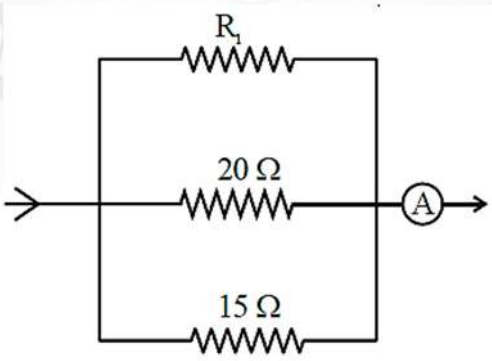
47. Hydrogen atom is bombarded with electrons accelerated through a potential difference of V, which causes excitation of hydrogen atoms. If the experiment is being performed at T = 0 K, the minimum potential difference needed to observe any Balmer series lines in the emission spectra will be $\frac{\alpha}{10}$V , where α=_____

48. A horizontal straight wire 5 m long extending from east to west falling freely at right angle to horizontal component of earth's magnetic field $0.60\times10^{-4}\,Wbm^{-2}$. The instantaneous value of emf induced in the wire when its velocity is $10ms^{-1}$ is _____ $\times10^{-3}$V .

49. In the given figure, the charge stored in 6μF capacitor, when points A and B are joined by a connecting wire is _____μC.



50. In the given circuit, the current flowing through the resistance 20Ω is 0.3 A, while the ammeter reads 0.9 A. The value of R_1 is _____Ω



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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. Which of the following options are correct for the reaction

$$2\left[Au(CN)_2\right]^-(aq) + Zn(s) \rightarrow 2Au(s) + \left[Zn(CN)_4\right]^{2-}(aq)$$

A. Redox reaction

B. Displacement Reaction.

C. Decomposition reaction.

D. Combination reaction

Choose the correct answer from the options given below :

1) A and D only **2)** A and B only **3)** A only **4)** C and D only

52. Match List – I with List – II.

	List – I **Name of reaction**		**List – II** **Reagent used**
A	Hell -Volhard - Zelinsky reaction	I	$NaOH + I_2$
B	Iodoform reaction	II	$(i)\ CrO_2Cl_2, CS_2\ \ (ii)\ H_2O$
C	Etard reaction	III	(i) Br_2/red phosphorus (ii) H_2O
D	Gatterman - Koch reaction	IV	CO, HCl, anhyd $AlCl_3$

Choose the correct answer from the options given below.

1) A - III, B - II, C - I, D - IV **2)** A - III, B - I, C - II, D - IV

3) A–I,B–II,C–III,D–IV **4)** A - III, B - I, C - IV, D - II

53. The standard electrode potential of M^+/M in aqueous solution does not depend on

1) Ionisation of a solid metal atom **2)** Ionisation of a gaseous metal atom

3) Sublimation of a solid metal **4)** Hydration of a gaseous metal ion

54. Given below are two statements, one is labeled as Assertion A and the other is labeled as Reason R

Assertion A : Loss of electron from hydrogen atom results in nucleus of $\sim 1.5 \times 10^{-3}$ pm size.

Reason R : Proton (H^+) always exists in combined form

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In the light of the above statements , Choose the most appropriate answer from the options given below.

1) Both (A) and (R) are correct but (R) is the correct explanation of (A).

2) Both (A) and (R) are correct but (R) is NOT the correct explanation of (A)

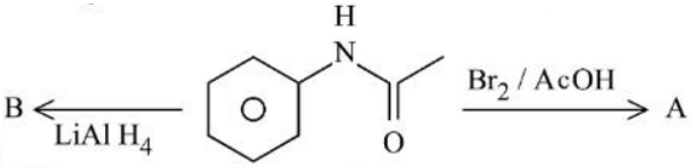
3) (A) is correct but (R) is not correct

4) (A) is not correct but (R) is correct

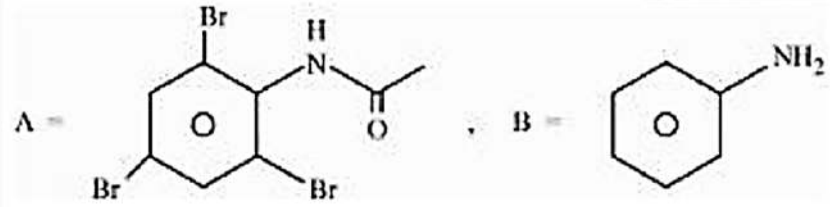
55. If the radius of the first orbit of hydrogen atom is a_0, then de Broglie's wave length of electron in 3rd orbit is

1) $3\pi a_0$ **2)** $\pi a_0/6$ **3)** $\pi a_0/3$ **4)** $6\pi a_0$

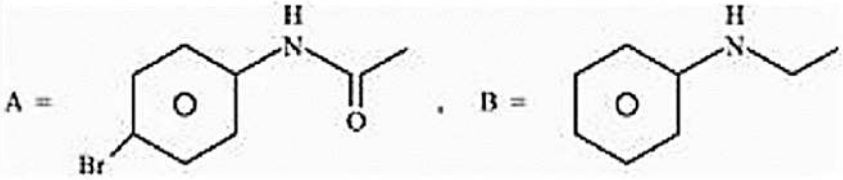
56. The major product A and B from the following reactions are



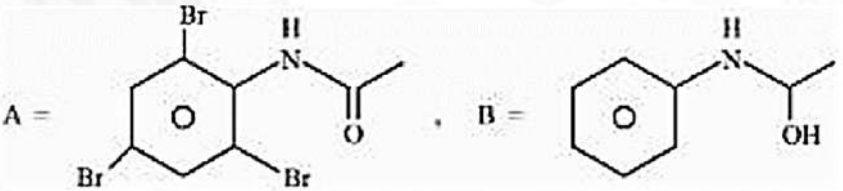
1)



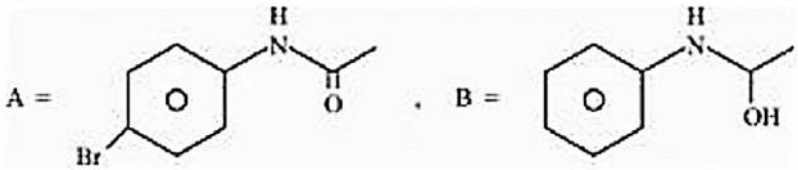
2)



3)



4)



57. Given below are two statements, one is labeled as Assertion A and the other is labeled as Reason R.

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Assertion A : The spin only magnetic moment value for $[Fe(CN)_6]^{3-}$ is 1.74BM, whereas for $\left[Fe(H_2O)_6\right]^{3+}$ is 5.92 BM.

Reason R : In both complexes, Fe is present in +3 oxidation state.

In the light of the above statements, choose the correct answer from the options given below.

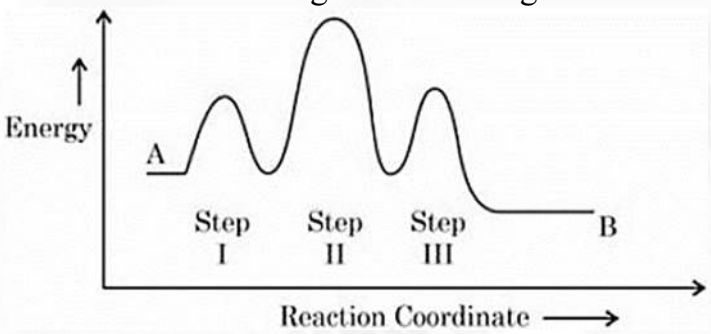
1) Both (A) and (R) are true and (R) is the correct explanation of (A).

2) Both (A) and (R) are true and (R) is not the correct explanation of (A).

3) (A) is true but (R) is false.

4) (A) is false but (R) is true.

58. Consider the following reaction that goes from A to B in three steps as shown below:



Choose the correct option

	Number of Intermediates	Number of activated complexes	Rate determining step
1)	2	3	II
2)	2	3	III
3)	3	2	II
4)	2	3	I

59. Strong reducing and oxidizing agents among the following respectively, are

1) Eu^{2+} and Ce^{4+} **2)** Ce^{4+} and Tb^{4+} **3)** Ce^{3+} and Ce^{4+} **4)** Ce^{4+} and Eu^{2+}

60. Match List I with List II.

	List-I Vitamin		List-II Deficiency Disease
A.	Vitamin A	I.	Beri -Beri
B.	Thiamine	II.	Cheilosis
C.	Ascorbic acid	III.	Xeropthalmia
D.	Riboflavin	IV.	Scurvy

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Choose the correct answer from the options given below :

1) A-III, B-II, C-IV, D-I **2)** A–IV, B–II, C–III, D–I

3) A-III, B-I, C-IV, D-II **4)** A–IV, B–I, C–III, D–II

61. For the reaction

$$RCH_2Br + I^- \xrightarrow{\text{Acetone}} \underset{\text{major}}{RCH_2I} + Br^-$$

The correct statement is

1) The transition state formed in the above reaction is less polar than the localized anion

2) The solvent used in the reaction solvates the ions formed in rate determining step

3) Br^- can act as competing nucleophile

4) The reaction can occur in acetic acid also

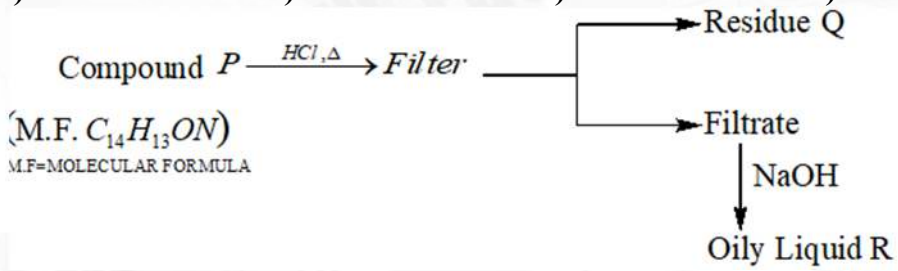
62. For a concentrated solution of a weak electrolyte (K_{eq} = equilibrium constant) A_2B_3 of concentration 'c', the degree of dissociation 'α' is

1) $\left(\frac{K_{eq}}{5c^4}\right)^{\frac{1}{5}}$ **2)** $\left(\frac{K_{eq}}{108c^4}\right)^{\frac{1}{5}}$ **3)** $\left(\frac{K_{eq}}{6c^5}\right)^{\frac{1}{5}}$ **4)** $\left(\frac{K_{eq}}{25c^2}\right)^{\frac{1}{5}}$

63. The difference between electron gain enthalpies will be maximum between

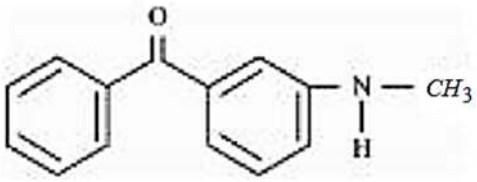
1) Ar and Cl **2)** Ne and F **3)** Ar and F **4)** Ne and Cl

64.

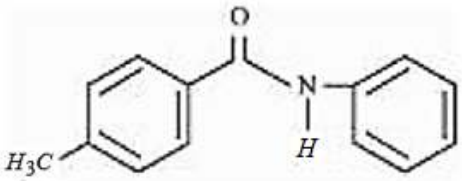


Compound P is neutral Q gives effervescence with $NaHCO_3$ while R reacts with Hinsbergs reagent to give solid soluble in NaOH. Compound P is

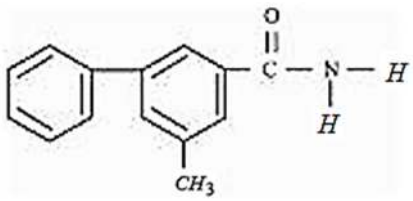
1)



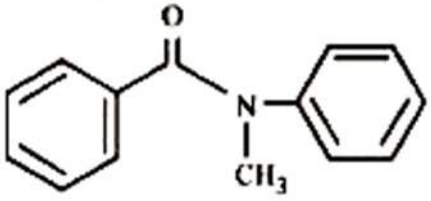
2)



3)



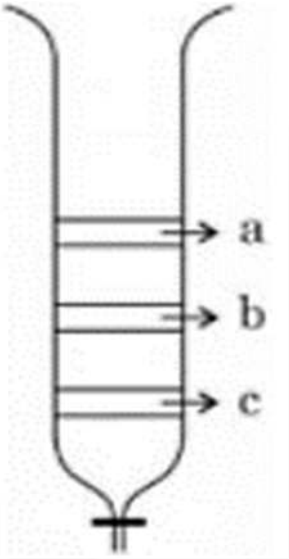
4)



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65. From the figure of column chromatography given below, identify incorrect statements.



A) Compound 'c' is more polar than 'a' and 'b'

B) Compound 'a' is least polar

C) Compound 'b' comes out of the column before 'c' and after 'a'

D) Compound 'a' spends more time in the column

Choose the correct answer from the options given below:

1) A B and D only **2)** A, B and C only **3)** B and D only **4)** B, C and D only

66. Match List –I with List – II

List - I Enzymatic Reaction		List - II Enzyme	
A	Sucrose → Glucose and Fructose	I	Zymase
B	Glucose → ethyl alcohol and CO_2	II	Pepsin
C	Starch → Maltose	III	Invertase
D	Proteins → Amino acids	IV	Diastase

Choose the correct answer from the options given below:

1) A–I, B–II, C–IV, D–III **2)** A - III, B - I, C - IV, D - II

3) A–I, B–IV, C–III, D–II **4)** A - III, B - I, C - II, D - IV

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67. Match List –I with List -II

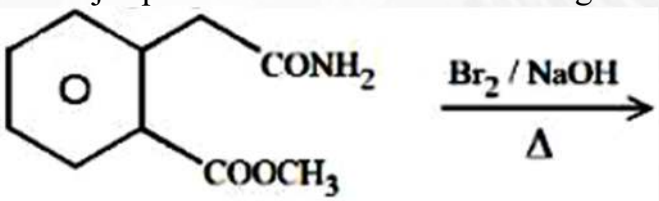
List - I Element detected		List - II Reagent used / Product formed	
A	Nitrogen	I	$Na_2[Fe(CN)_5NO]$
B	Sulphur	II	$AgNO_3$
C	Phosphorous	III	$Fe_4[Fe(CN)_6]_3$
D	Halogen	IV	$(NH_4)_2MoO_4$

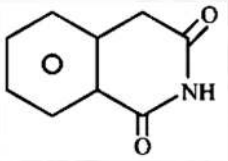
Choose the correct answer from the options given below

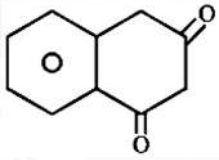
1) A–II, B–IV, C–I, D - III **2)** A - IV, B - II, C - I, D - III

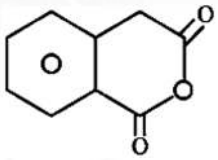
3) A - III, B - I, C - IV, D - II **4)** A–II, B–I, C–IV, D–III

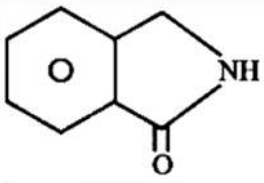
68. The major product formed in the following reaction is



1) 

2) 

3) 

4) 

69. The volume of 0.02 M aqueous HBr required to neutralize 10.0 mL, of 0.01 M aqueous $Ba(OH)_2$ is (Assume complete neutralization)

1) 5.0 mL **2)** 10.0 mL **3)** 2.5 mL **4)** 7.5 mL

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70. Match List – I with List – II

List - I Oxide		List - II Type of bond	
A	N_2O_4	I	1N = O bond
B	NO_2	II	1N – O – N bond
C	N_2O_5	III	1N – N bond
D	N_2O	IV	1N = N / N ≡ N bond

Choose the correct answers from the options given below:

1) A–II, B–I, C–III, D–IV **2)** A - III, B - I, C - II, D - IV

3) A - II, B - IV, C - III, D - I **4)** A - III, B - I, C-IV, D-II

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. Mass of Urea (NH_2CONH_2) required to be dissolved in 1000 g of water in order to reduce the vapour pressure of water by 25% is _____ g (Nearest integer)

Given : Molar mass of N, C, O and H are 14,12,16 and 1 g mol^{-1} respectively.

72. The value of log K for the reaction A⇌B at 298 K is _____ (Nearest integer)

Given : $\Delta H^\circ = -54.07 kJmol^{-1}$

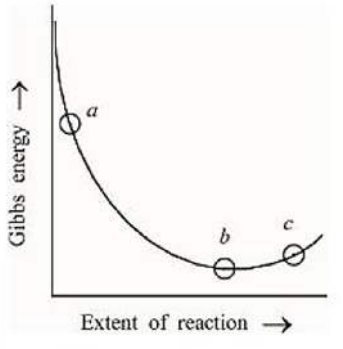
$\Delta S^\circ = 10JK^{-1}mol^{-1}$

(Take 2.303×8.314×298=5705)

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73. Consider the graph of Gibbs free energy G vs Extent of reaction. The number of statement/s from the following which are true with respect to points (a), (b) and (c) is



_____

A. Reaction is spontaneous at (a) and (b)

B. Reaction is at equilibrium at point (b) and non-spontaneous at point (c)

C. Reaction is spontaneous at (a) and non-spontaneous at (c)

D. Reaction is non – spontaneous at (a) and (b)

74. Number of ambidentate ligands in a representative metal complex $\left[M(en)(SCN)_4\right]$ is _____[en = ethylenediamine]

75. Number of bromo derivatives obtained on treating ethane with excess of Br_2 in diffused sunlight is

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29	113	205	745	4,212

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